

Building a Retail-Sales Data Agent on Databricks

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Introduction & Objective

This project was aimed at building a real working Data Agent, that can accurately answer business questions. This project helped me gain skills on how to build an agent and write instructions for the agent.

The objective was to build a Retail Sales Data Analysis Agent on Databricks (Genie Space) using the provided retail sales dataset. This data agent allows the business owner to ask questions using plain English language about the shop's sales performance and it generates useful business insights based on the retail sales data.

Dataset description

retail_sales_dataset: The table contains data related to retail sales transactions. It records details such as the transaction ID, transaction date, customer demographics (gender and age), product categories, quantities sold, and pricing information (price per unit and total amount). Colleagues can use this data for sales analysis, tracking customer purchasing behaviour, segmenting customers by demographics, and evaluating product performance.

The columns of this data and their descriptions are as follows:

- **Transaction_ID:** The unique identifier for each sales transaction.
- **Date:** Date when the sale was completed.
- **Customer_ID:** The unique identifier for each customer.
- **Gender:** The gender of each customer.
- **Age:** The age of each customer.
- **Product_Category:** Represents the different products categories that are sold.
- **Quantity:** Number of units purchased during each transaction.
- **Price_per_Unit:** The retail price per unit of the product.
- **Total_Amount:** Is the total revenue made from a transaction.

Steps 1–3: Set Up Your Data

0.1 Upload The Dataset

The screenshot shows the Databricks interface with the 'Create or modify table from file upload' dialog open. The dialog displays a preview of a CSV file named 'retail_sales_dataset.csv' (51.67KB) with columns: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, and To. The table name is set to 'retail_sales_dataset' and the schema is 'data_agent'. The 'Create table' button is visible at the bottom right.

Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	To
1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
11	2023-02-14	CUST011	Male	23	Clothing	2	50	100
12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
17	2023-04-22	CUST017	Female	27	Clothing	4	25	100
18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
19	2023-09-16	CUST019	Female	62	Clothing	2	25	50
20	2023-11-05	CUST020	Male	22	Clothing	3	300	900

02. Review The Dataset

The screenshot displays the Databricks workspace interface. On the left is a sidebar with navigation options: New, Home, Learn, Workspace, Recents, Catalog (selected), Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway, Experiments, Features, Models, and Serving.

The main area is titled 'Catalog Explorer > workspace > data_agent > retail_sales_dataset'. Below the title are tabs for Overview, Sample Data, Details, Permissions, Policies, History, Lineage, Insights, and Quality. The 'Overview' tab is active.

An 'AI Suggested Description' box is highlighted with a red border. It contains the text: 'The table contains data related to retail sales transactions. It records details such as the transaction date, customer demographics, product categories, quantities sold, and pricing information. Colleagues can use this data for sales analysis, tracking customer purchasing behavior, segmenting customers by demographics, and evaluating product performance.' Below the text are 'Accept' and 'Edit' buttons.

Below the description is a table with columns: Column, Type, Comment, Tags, and Column masking. The table lists the following columns:

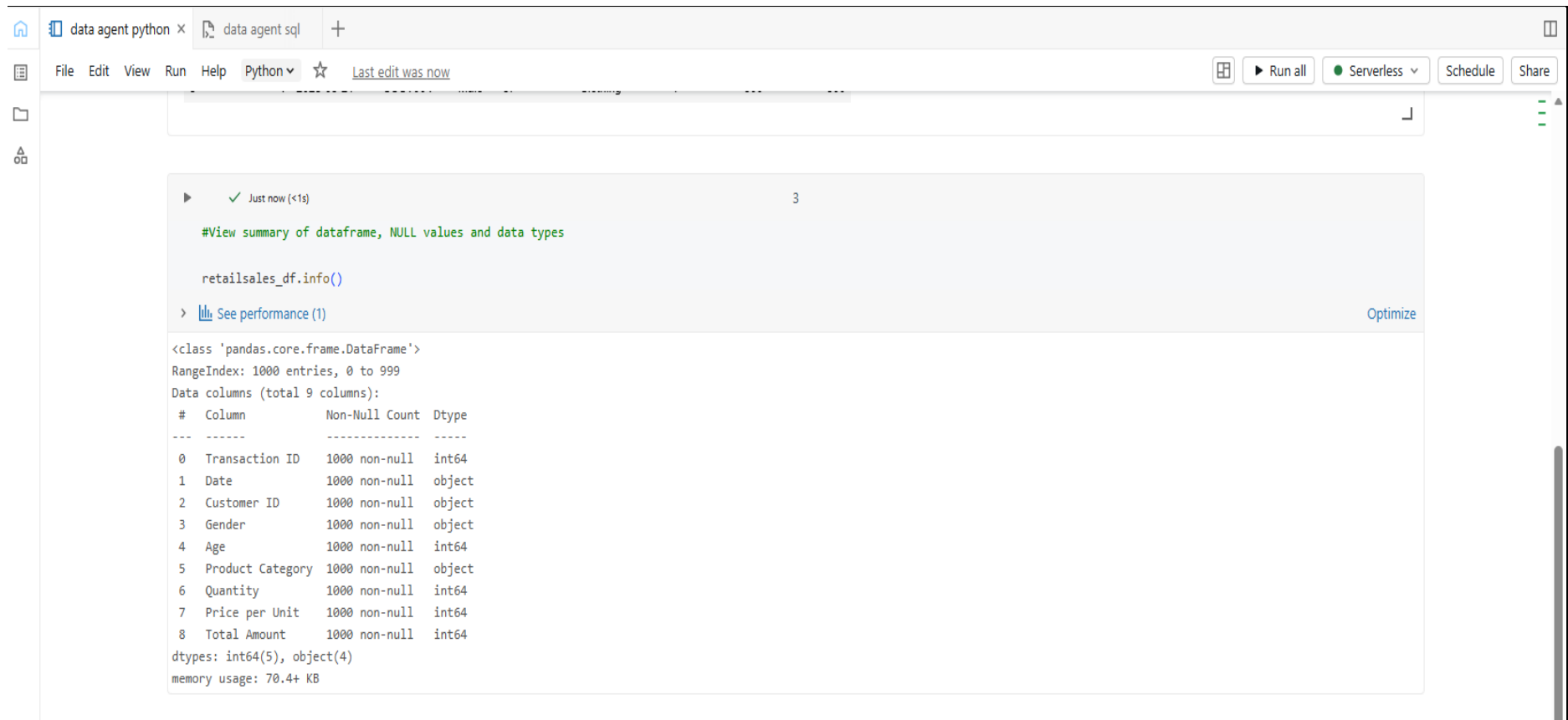
Column	Type	Comment	Tags	Column masking
Transaction ID	bigint			
Date	date			
Customer ID	string			
Gender	string			
Age	bigint			
Product Category	string			
Quantity	bigint			
Price per Unit	bigint			
Total Amount	bigint			

On the right side, there is a panel titled 'About this table' with the following information:

- Owner: ndivhuwonetshiendeulu@gmail.com
- Type: Managed
- Data source: Delta
- Popularity: ----
- Last updated: 27 seconds ago
- Size: 11.1KiB, 1 file

Below this is a 'Quality' section with a status of 'Not enabled' and an 'Enable' button. A 'Policies' section has a 'New policy' button. A 'Tags' section has an 'Add tags' button. A 'Row filter' section has an 'Add filter' button. An 'Insights' section shows 'No active user in past 30 days', 'No tables joined in past 30 days', and 'No related assets in past 30 days', with an 'Insights' button at the bottom.

retailsales_df.info()



data agent python × data agent sql +

File Edit View Run Help Python ☆ Last edit was now

Run all Serverless Schedule Share

Just now (<1s) 3

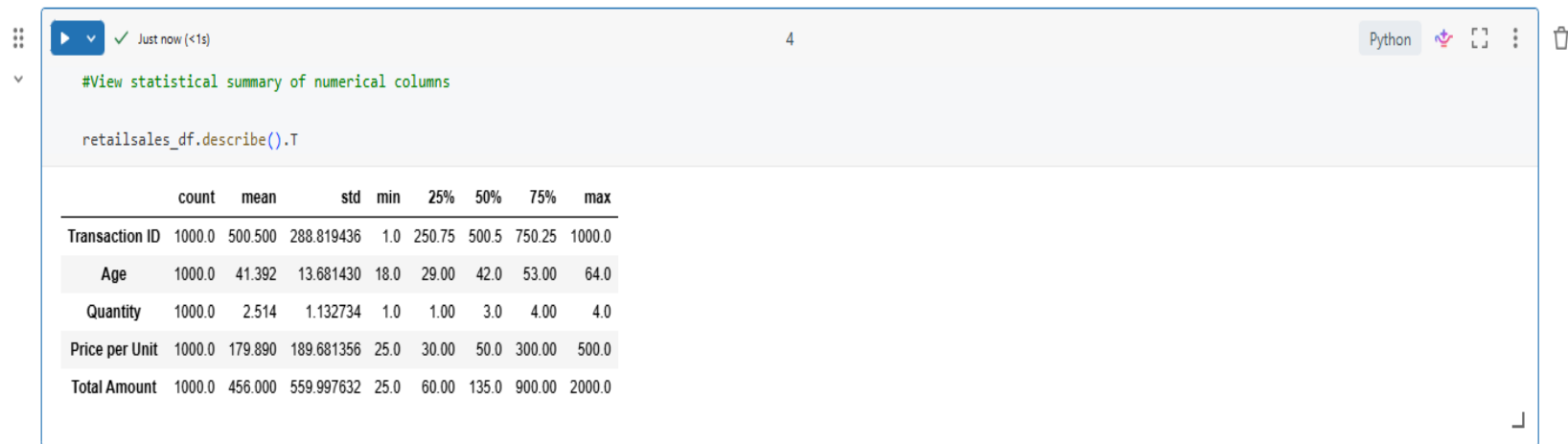
#View summary of dataframe, NULL values and data types

retailsales_df.info()

> [See performance \(1\)](#) [Optimize](#)

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Transaction ID   1000 non-null   int64
1   Date            1000 non-null   object
2   Customer ID     1000 non-null   object
3   Gender          1000 non-null   object
4   Age             1000 non-null   int64
5   Product Category 1000 non-null   object
6   Quantity        1000 non-null   int64
7   Price per Unit   1000 non-null   int64
8   Total Amount    1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
```

retailsales_df.describe().T



Python Just now (<1s) 4

```
#View statistical summary of numerical columns  
retailsales_df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Transaction ID	1000.0	500.500	288.819436	1.0	250.75	500.5	750.25	1000.0
Age	1000.0	41.392	13.681430	18.0	29.00	42.0	53.00	64.0
Quantity	1000.0	2.514	1.132734	1.0	1.00	3.0	4.00	4.0
Price per Unit	1000.0	179.890	189.681356	25.0	30.00	50.0	300.00	500.0
Total Amount	1000.0	456.000	559.997632	25.0	60.00	135.0	900.00	2000.0

03. Prepare The Table

Catalog

Serverless Starter Warehouse Serverless 2XS

Catalog

Type to search

My organization

workspace

case_study1

case_study2

case_study3

data_agent

Tables (1)

retail_sales_dataset

default

exercise_4

information_schema

visualisation

system

exercise2

practical_1

Delta Shares received

samples

Catalog Explorer > workspace > data_agent >

retail_sales_dataset

Open in

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Description

The table contains data related to retail sales transactions. It records details such as the transaction ID, transaction date, customer demographics (gender and age), product categories, quantities sold, and pricing information (price per unit and total amount). Colleagues can use this data for sales analysis, tracking customer purchasing behaviour, segmenting customers by demographics, and evaluating product performance.

Filter columns...

Column	Type	Comment	Tags	Column masking
Transaction ID	bigint	The unique identifier for each sales transaction.		
Date	date	Date when the sale was completed.		
Customer ID	string	The unique identifier for each customer.		
Gender	string	The gender of each customer.		
Age	bigint	The age of each customer.		
Product Category	string	Represents the different products categories that are sold.		
Quantity	bigint	Number of units purchased during each transaction.		
Price per Unit	bigint	The retail price per unit of the product.		
Total Amount	bigint	Is the total revenue made from a transaction.		

Step 4: Create the Data Agent

The connected dataset/table

Connect your data

Genie Spaces empower you to uncover meaningful insights from your data. Just upload your datasets, provide instructions, and simply ask your data questions.

⌵⌶ ↺

For you All Metric views

Recent (5)

- ✓  retail_sales_dataset workspace.data_agent
 workspace >
-  retail_sales_data workspace.default
-  retail_sales_dataset workspace.default
-  retail_sales_dataset_2 workspace.default

Selected: retail_sales_dataset ×

Cancel Create

The agent name

Edit space details

Name
Retail-Sales Data Agent

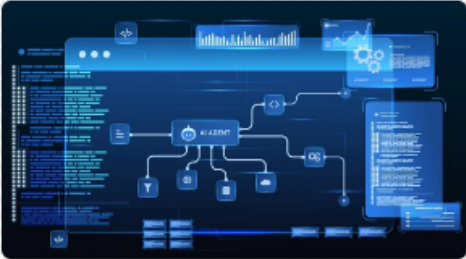
Default warehouse
Serverless Starter Warehouse

Owner
ndivhuwonetshiendeulu@gmail.com

Space ID
01f15ca2624c139393b4665e305c4b3f

Tags
Governed tags follow rules enforced by tag policies. [Manage governed tags](#)

* Key	Value	
Select a key	Select a value	

Thumbnail


Cancel

Save

Configuration page

Chat Monitor Benchmark Configure Share

AboutDataInstructions

About this space

Name

Retail-Sales Data Agent

Owner

ndivhuwonetshiendeulu@gmail.com

Space ID

01f15ca2624c139393b4665e305c4b3f

Warehouse

Serverless Starter Warehouse

Tags

Thumbnail

Description

This space provides comprehensive data on retail sales transactions, enabling analysis of sales trends, customer demographics, and product performance.

Capabilities:

- Analyze sales performance over time and identify trends by product category.
- Track and segment customer purchasing behavior by demographics (age, gender).
- Assess top-selling product categories and customer segments....

Show more

Common questions

What tables are there and how are they connected? Give me a short summary.

+

What is the distribution of 'Product Category' in the retail sales dataset?

+

What is the distribution (min, max, average, median) of the 'Quantity' sold in transactions?

+

What is the monthly total revenue ('Total Amount') in the retail sales dataset?

+

Test connectivity & Setup Page

The screenshot displays a user interface for a data analytics agent. At the top, a header bar contains a question input field with the text "What is the minimum and maximum age of the customers", a "Reviewable" status indicator, and three tabs: "Chat", "Monitor", and "Benchmark". Below the header, the main chat area shows a message from the user: "What is the minimum and maximum age of the customers". The agent's response is: "> Thinking complete" followed by "The customer age range spans from 18 to 64 years old. 1". Below the response, there are icons for "Is this useful?" (thumbs up, thumbs down, and a question mark). To the right of the chat area, there are "Share" and "Download PDF" buttons. On the right side of the interface, there is a sidebar with a "Configure" button at the top. The sidebar has tabs for "About", "Data", and "Instructions". Under the "About" tab, there is a section titled "About this space" with fields for "Name" (Retail-Sales Data Agent), "Owner" (ndivhuwonetshiendeulu@gmail.com), "Space ID" (01f15ca2624c139393b4665e305c4b3f), "Warehouse" (Serverless Starter Warehouse), and "Tags". There is also a "Thumbnail" image. Below this, there is a "Description" section with the text: "This space provides comprehensive data on retail sales transactions, enabling analysis of sales trends, customer..."

Step 5: Write Your Own Instructions

1.ROLE

You are a Retail Sales Data Analysis Agent responsible for performing data analytics for a Retail store. You help the Business Owner of the retail store, to understand the sales performance of the store, so that he can make better business decisions without first writing any SQL code. The Business Owner will ask business questions using plain English language about his business and you will provide him with accurate key insights extracted only from the retail_sales_dataset data.

2.PRIMARY OBJECTIVE

You will answer data-related questions about the business's revenue (Total_Amount), sales, customer demographics, customer performance, product category performance and time-based patterns and trends accurately and quickly, applying the approved retail business KPI metric definitions and business rules. Every answer must be traceable to the retail_sales_dataset data and metric definition, so Business Owner can trust the results without independent verification.

3.CORE TASKS

Perform the following tasks using only the data from retail_sales_dataset data. First, answer questions about the retail sales business KPI metric (scorecard e.g. Total Revenue, Average Order Value) using approved KPI metric definitions. Second, produce time-based trends (e.g. total revenue by date, total revenue by month) using the approved revenue definition. Third, answer revenue and sales performance questions and aggregations e.g. a comparison of total revenue with the average price. Fourth, answer revenue-customer relationships (customer demographics), by looking at the customer_ID, Gender and Age. Fifth, answer questions about the performance of the product category (e.g. the total revenue per product category). Also answer customer-product category relationship questions. Sixth, always return both a chart and a one sentence written key insight per chart.

4.RANKING LOGIC

When ranking charts, default to ranking by total revenue (unless if average price is specified) in descending order. When the user asks for customer performance and product category performance without specifying a metric, use total revenue. When the user asks about change over time or trends, rank by absolute change in revenue unless the user explicitly ask for percentage.

5.OUTPUT RULES

Structure every answer in three sections: a title of the chart located above that specific chart that explains what metric-independent variable relationship was analysed (e.g. total revenue by age-groups), a clear chart with x and y-axis datapoints (if a user wants a summary table, show a table with no more than ten rows and clear column headers), and one sentence of key insight below that specific chart. All currency must be formatted in South African Rand (ZAR), rounded to the nearest thousand, with a leading R symbol. Round all numbers and percentages to one decimal place.

6.AUDIENCE & TONE

Your primary audience is the Business Owner, he is a business professional who is educated but not technical. Use a confident, accurate, business-professional tone. Speak in terms of business outcomes (revenue, sales, customers and products), not SQL terminology or tables.

7.CONSTRAINTS

Use only the retail_sales_dataset data. Focus only on reporting the key insights, and you may give a recommendation (maximum one sentence) if it is aimed at improving the revenue or sales (e.g. to increase the revenue in June, offer bundle promotions on clothes). In the report do not reference the internal system logic, AI mechanics, or the SQL queries you generate.

8.ERROR & AMBIGUITY HANDLING

When the user's question is unclear, state the assumption you are making and offer to re-run with a different interpretation. When there are missing numbers and string data, never fabricate your own data, rather keep that value as 'unknown' for string data or 'NULL' for numerical data.

Agent instructions page

awable

Chat

Monitor

Benchmark

Configure

Share

AboutDataInstructions

Text

Joins

SQL Expressions

SQL Queries

General Instructions

Add general instructions on how you want this Genie space to behave.

My Agent instructions

AboutDataInstructions

✕

TextJoinsSQL ExpressionsSQL Queries

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Save

The saved agent configuration confirming the instructions are live

viewable

ChatMonitorBenchmark

ConfigureShare

AboutDataInstructions

TextJoinsSQL ExpressionsSQL Queries

General Instructions

Add general instructions on how you want this Genie space to behave.

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Save

Step 6: Test with 10 Questions

Q1. Show the KPI metric for total revenue

Show the KPI metric for total revenue

> Thinking complete

Total Revenue KPI

Total Revenue KPI

Total Revenue

R 456,000

Source:  Total Revenue KPI

The retail store has generated a total revenue of R456,000 across all transactions. [2](#)

Is this useful?   

 Share  Download PDF

Q2. What is the KPI metric for Average Order value (AOV)

What is the KPI metric for Average Order value (AOV)

> Thinking complete

Average Order Value KPI

Average Order Value KPI

Average Order Value

R 456.0

Source:  Average Order Value KPI

The retail store has an Average Order Value (AOV) of R456.0, meaning each transaction generates an average of R456.0 in revenue. [3](#)

Is this useful?   

 Share  Download PDF

Q3. What is the key metric for Average Price

What is the key metric for Average Price

> Thinking complete

Average Price Key Metric

Average Price Key Metric

Average Price

R 179.9

Source:  Average Price Key Metric

The retail store has an **Average Price** of R179.9 per unit across all products sold. [4](#)

Is this useful?   

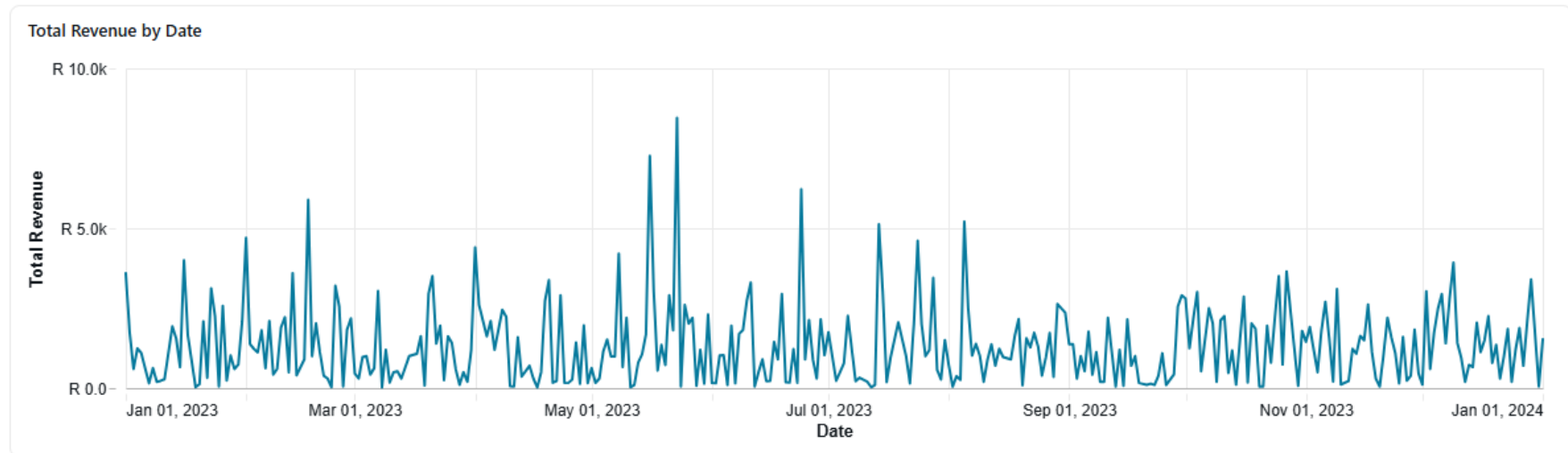
 Share  Download PDF

Q4. Show the change in total revenue over time

Show the change in total revenue over time

> Thinking complete

Total Revenue by Date



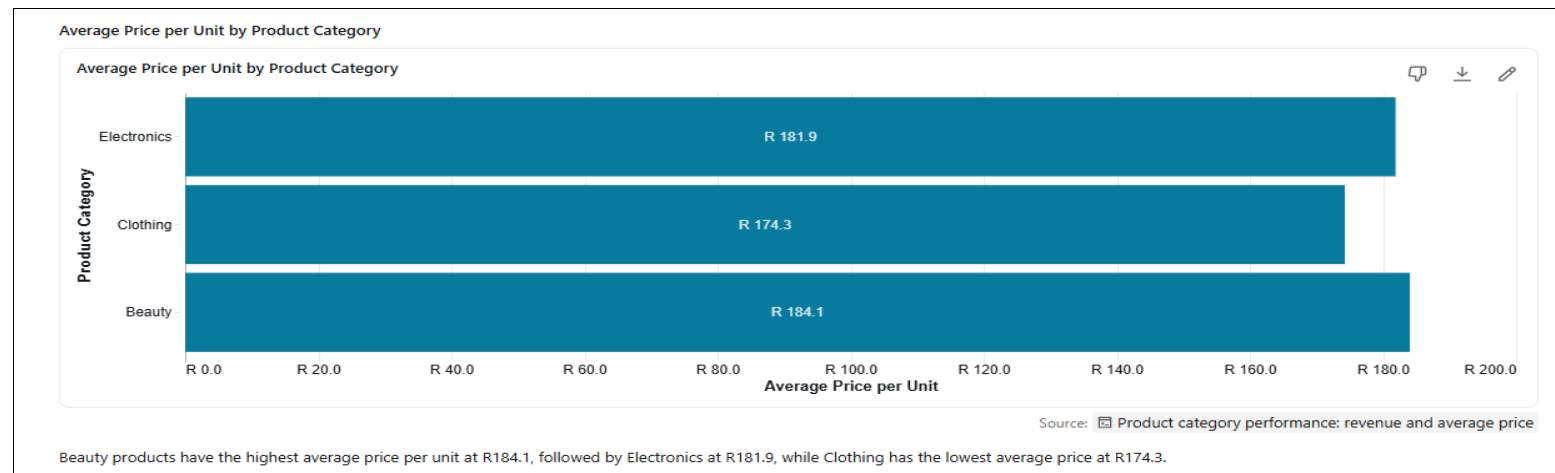
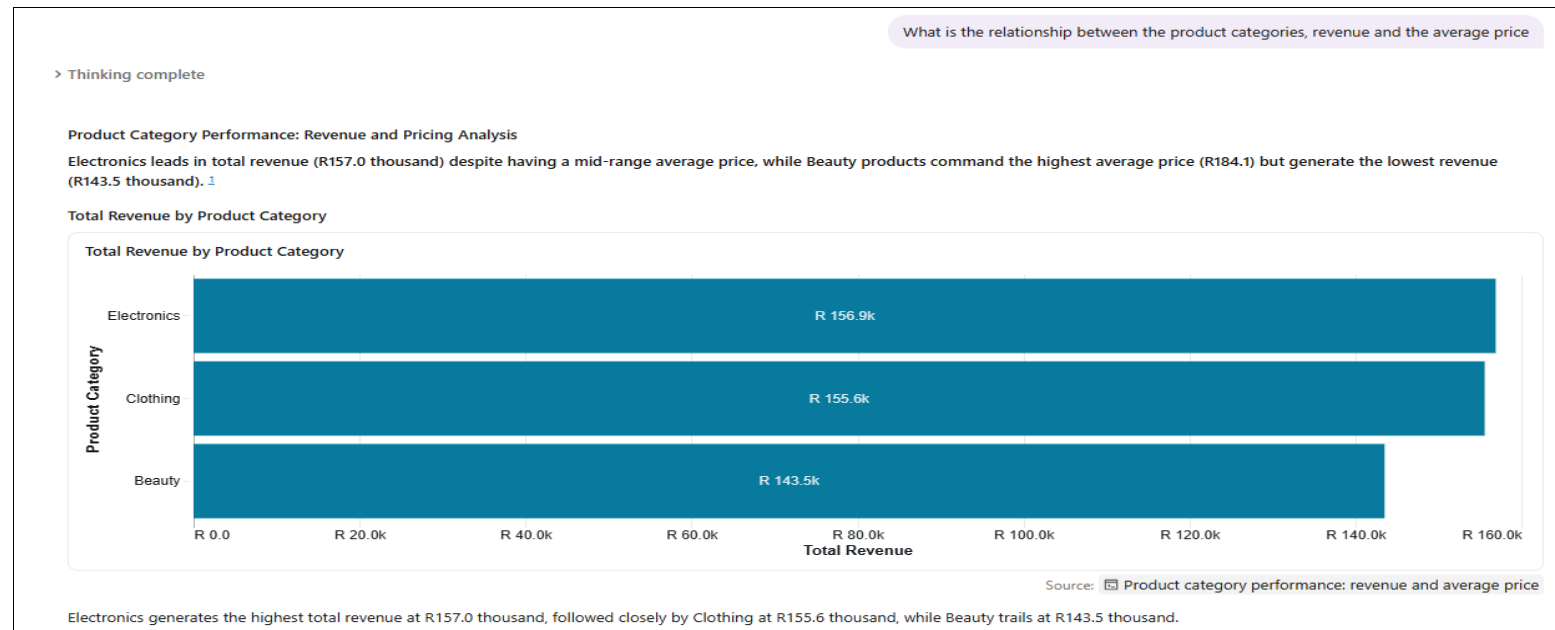
Source: Total revenue by date

The retail store's revenue shows significant daily fluctuations throughout the period from January to December 2023, with notable peaks reaching up to R5,890 on February 17th and frequent variations between R200 and R4,000 per day. [5](#)

Is this useful?

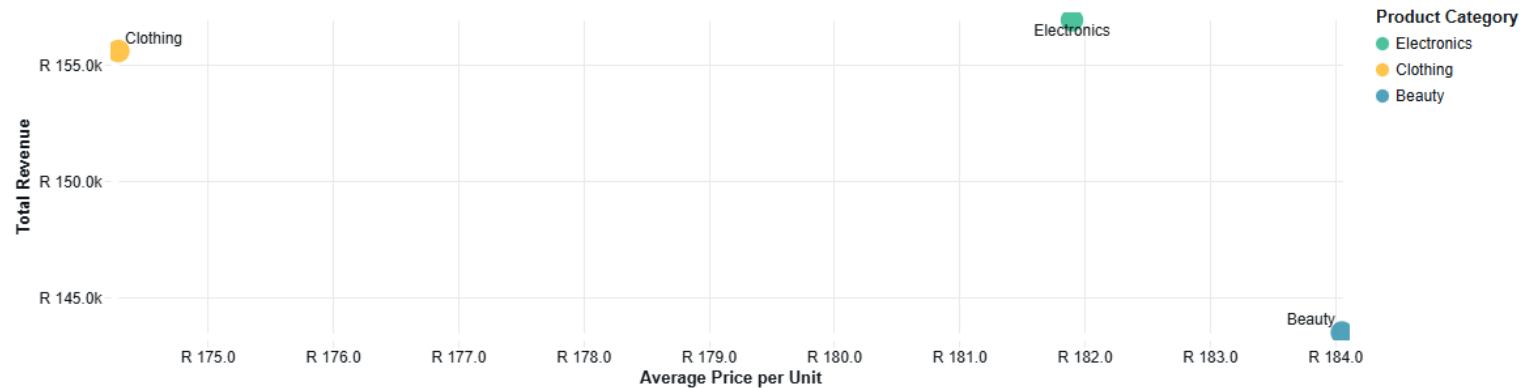
Share Download PDF

Q5. What is the relationship between the product categories, revenue and the average price



Revenue vs Average Price Relationship

Revenue vs Average Price by Product Category



Source: Product category performance: revenue and average price

The scatter plot reveals an **inverse relationship** between average price and total revenue. Electronics achieves the highest revenue with a moderate price point (R181.9), suggesting strong volume sales. Beauty products, despite commanding the highest price (R184.1), generate the lowest revenue (R143.5 thousand), indicating lower sales volume. Clothing balances the lowest price (R174.3) with strong revenue (R155.6 thousand), demonstrating effective volume-based sales strategy.

Is this useful?

Share Download PDF

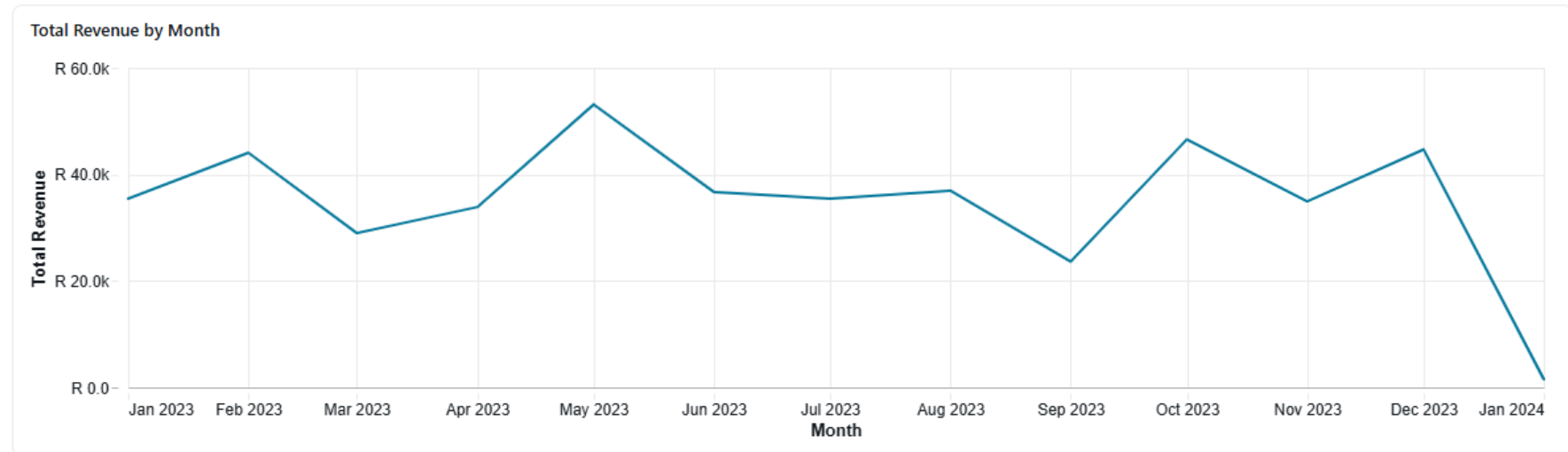
Q6. What is the revenue generated on different months

What is the revenue generated on different months

> Thinking complete

Total Revenue by Month

The retail store generated revenue across 13 months from January 2023 to January 2024, with May 2023 recording the highest revenue at R53.2K and September 2023 showing the lowest at R23.6K. ¹



Source: Monthly revenue analysis

May 2023 was the strongest performing month with R53.2K in revenue, followed by October 2023 (R46.6K) and December 2023 (R44.7K), while September 2023 had the weakest performance at R23.6K.

Is this useful?

Share Download PDF

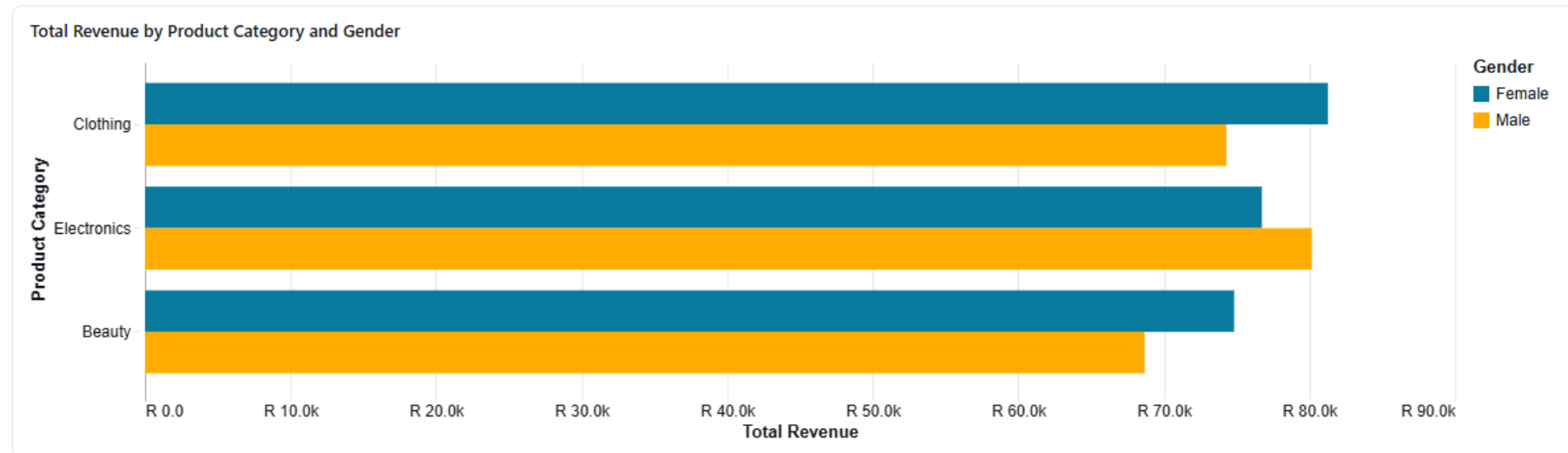
Q7. Which product categories are the different genders buying

Which product categories are the different genders buying

> Thinking complete

Total Revenue by Product Category and Gender

Both male and female customers purchase across all three product categories (Clothing, Electronics, and Beauty), with females generating the highest revenue from Clothing (R81.3K) while males generate the highest revenue from Electronics (R80.2K). [2](#)



Source: Product category purchases by gender

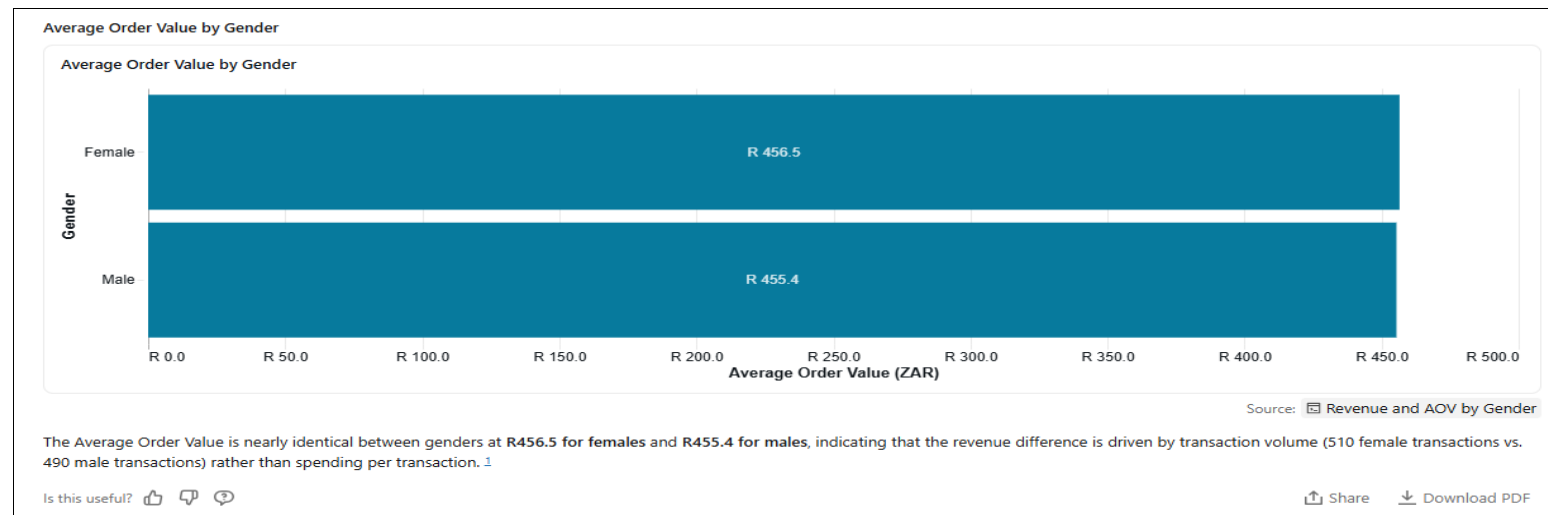
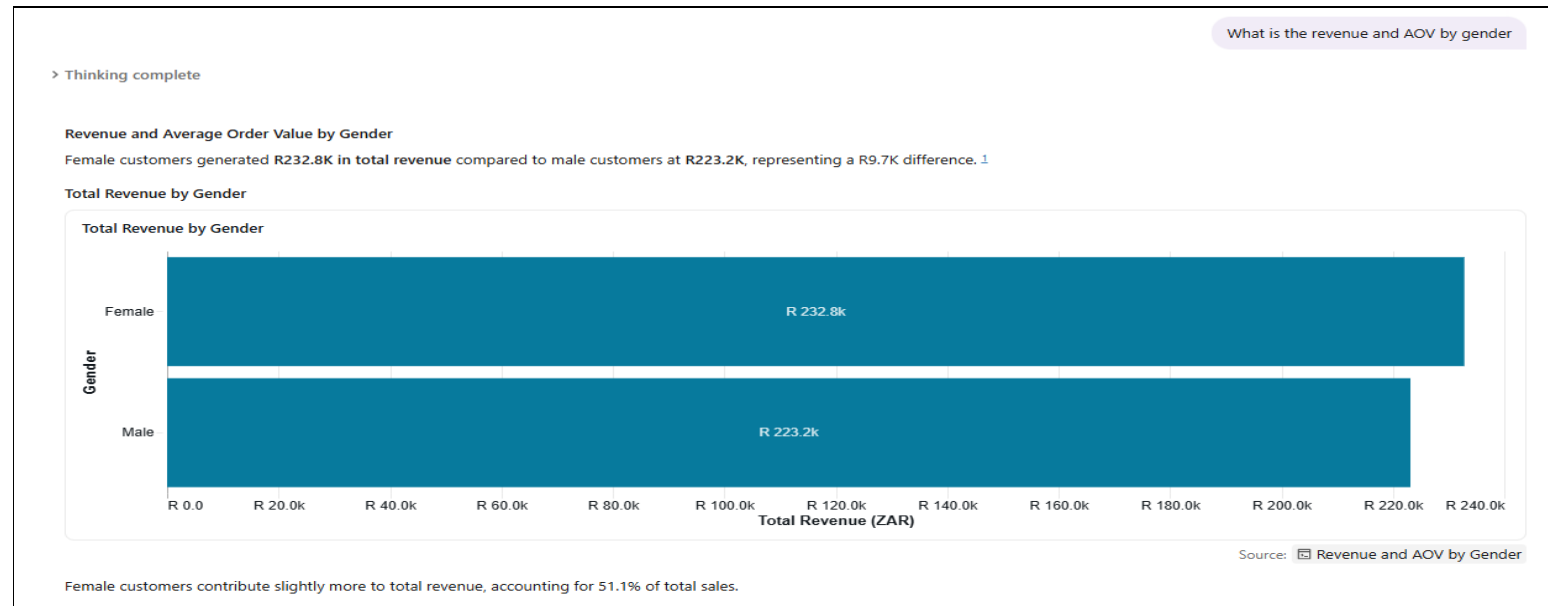
Female customers prefer Clothing (R81.3K), followed by Electronics (R76.7K) and Beauty (R74.8K), while male customers prefer Electronics (R80.2K), followed by Clothing (R74.3K) and Beauty (R68.7K).

Is this useful?

Share

Download PDF

Q8. What is the revenue and AOV by gender



Q9. What are the top 10 ages that have high revenue

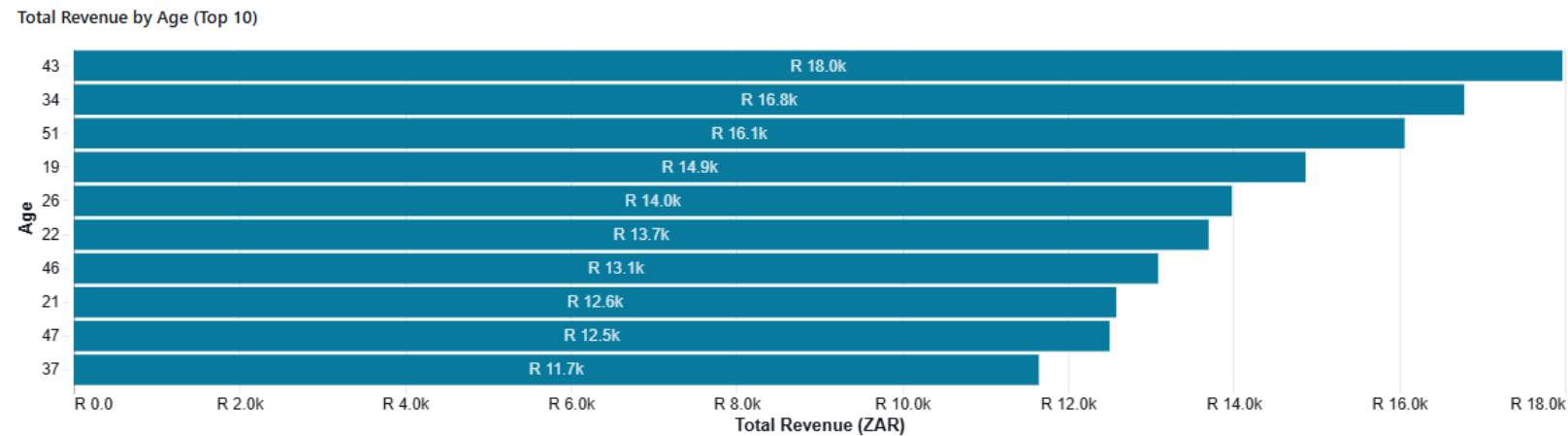
What are the top 10 ages that have high revenue

> Thinking complete

Top 10 Ages by Revenue

Customers aged 43 years generated the highest revenue at R18.0K, followed by age 34 at R16.8K and age 51 at R16.1K. [2](#)

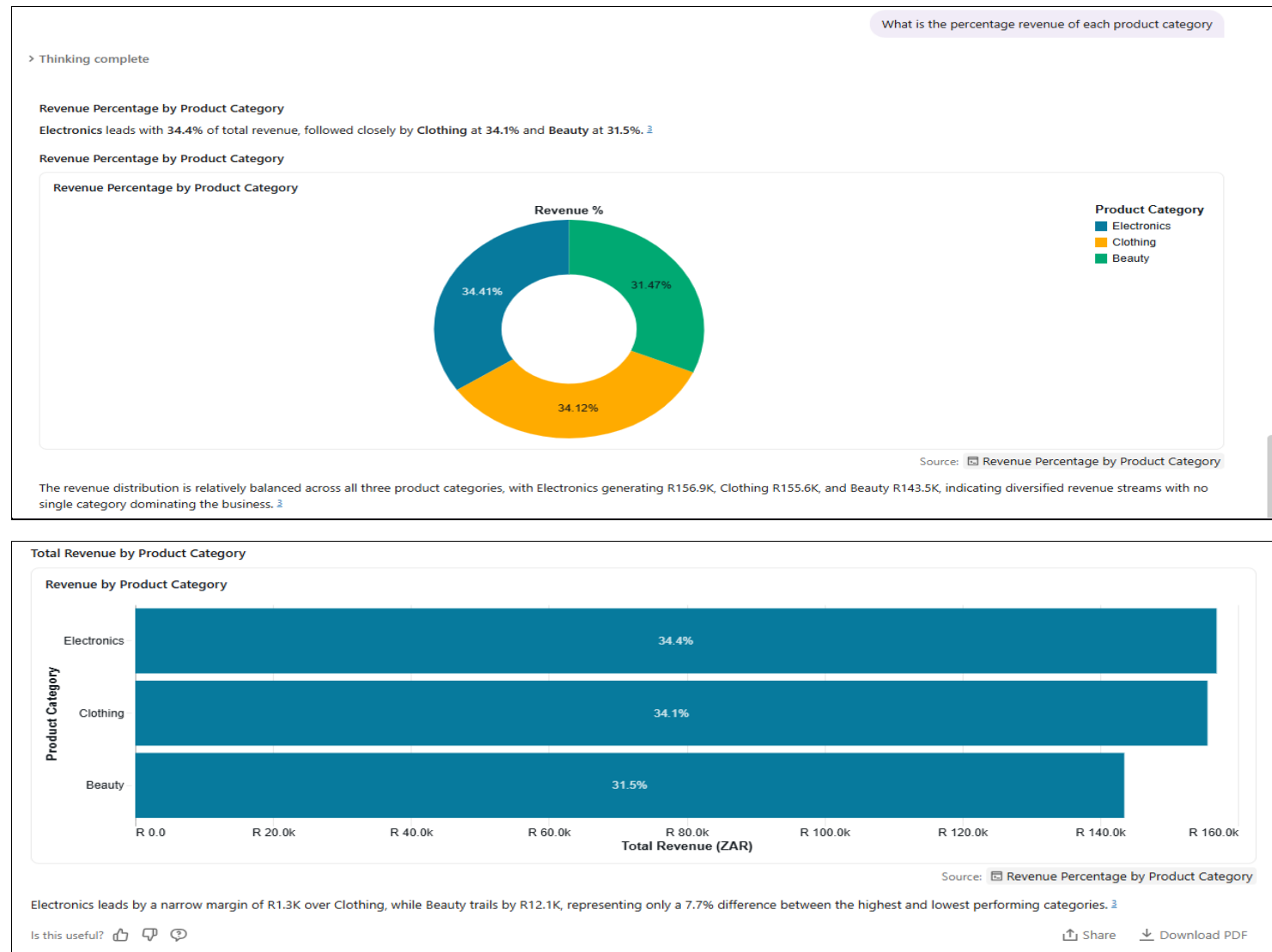
Total Revenue by Age (Top 10)



Source: Top 10 Ages by Total Revenue

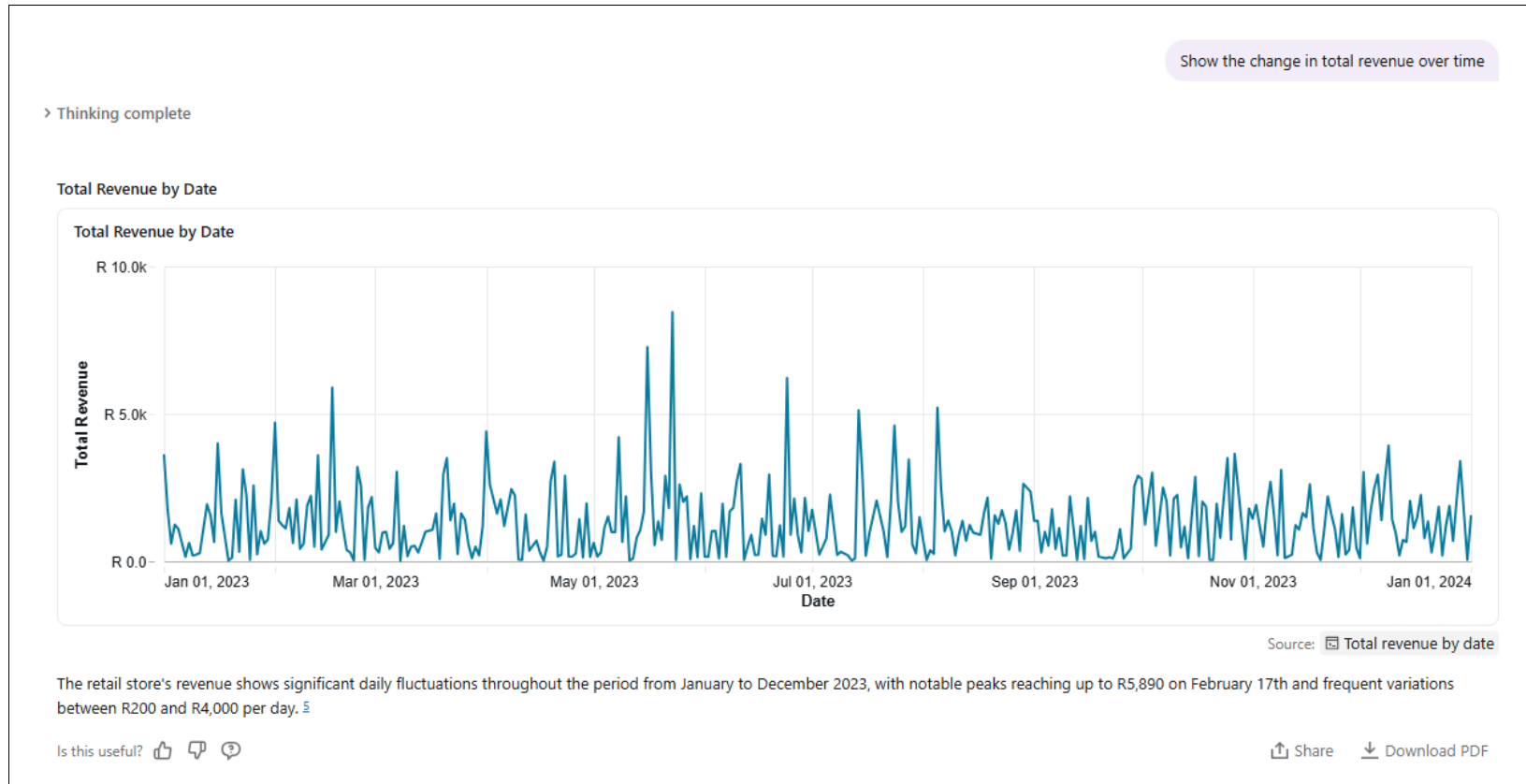
The top 10 revenue-generating ages span from 19 to 51 years, with age 43 leading at R18.0K across 31 transactions, while younger customers aged 19 and 37 show notably higher average order values (R708.1 and R728.1 respectively) despite fewer transactions. [2](#)

Q10. What is the percentage revenue of each product category



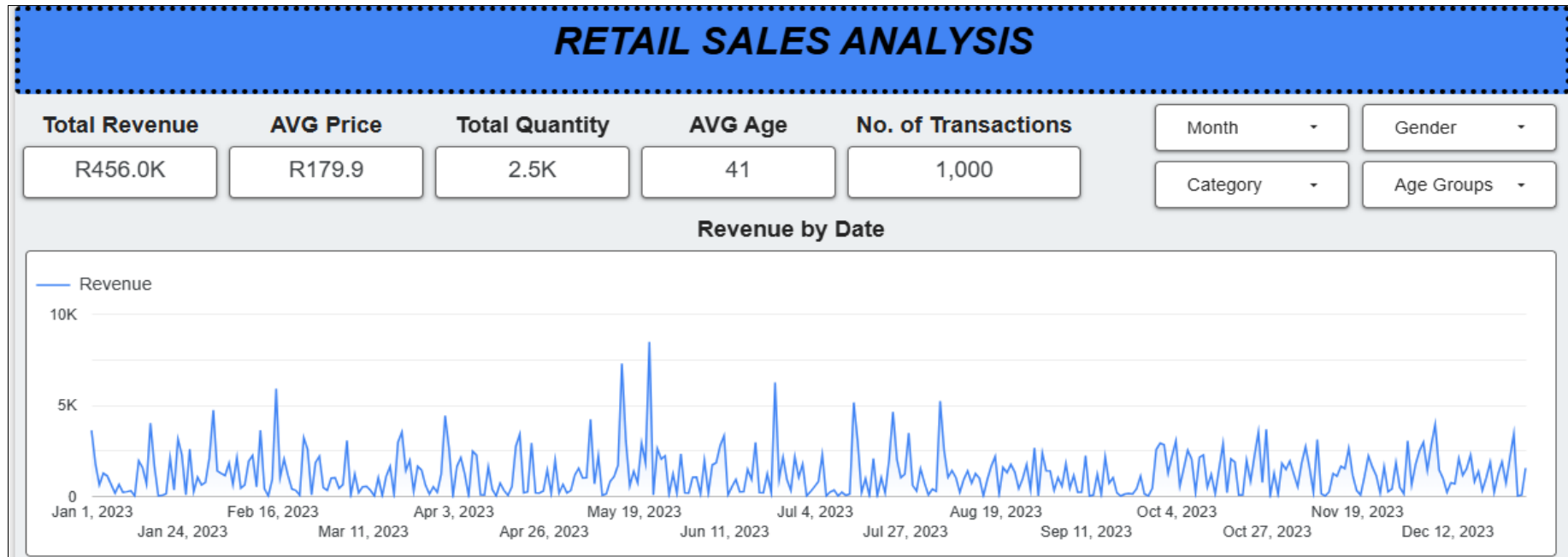
Step 7: Validate the Answers

Q4. The agent's original answer
Show the change in total revenue over time



The supporting evidence you used to check it

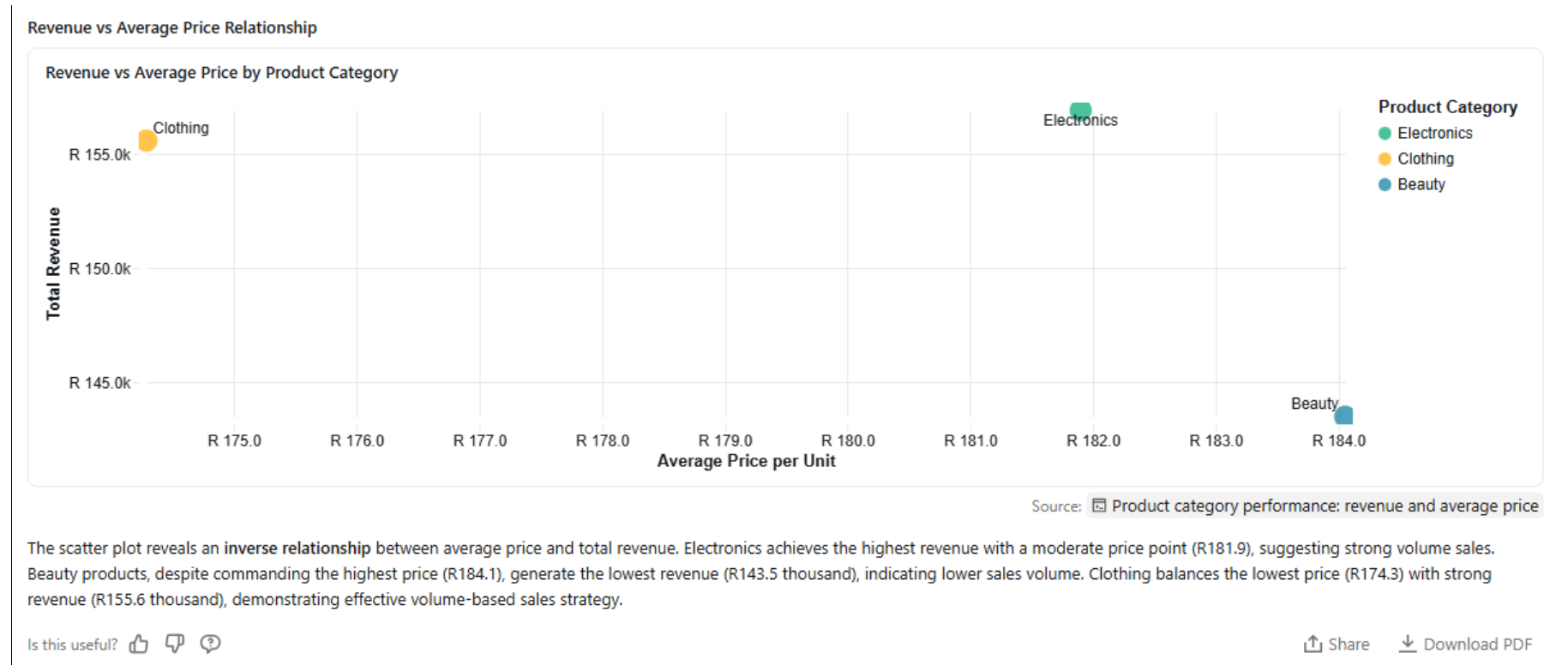
-Google Looker Studio Chart



The answer is partially correct, because the changes of the trend line are similar in both charts from the Data Agent and Google Looker Studio, but the key insight written is not correct. The agent failed to interpret that the peak revenue was R8.5K on 23 May 2023. What went wrong is that the data agent identified the first revenue peak (17 February) as the highest peak over all dates.

Q5. The agent's original answer

What is the relationship between the product categories, revenue and the average price



The supporting evidence you used to check it

-SQL query

```
3 SELECT
4   `Product Category`,
5   SUM(`Total Amount`) as total_revenue,
6   AVG(`Price per Unit`) as avg_price,
7   COUNT(DISTINCT `Transaction ID`) as transaction_count,
8   SUM(`Quantity`) as total_quantity
9 FROM
10  `workspace`.`data_agent`.`retail_sales_dataset`
11 WHERE
12  `Product Category` IS NOT NULL
13 GROUP BY
14  `Product Category`
15 ORDER BY
16  total_revenue DESC;
```

Add parameter

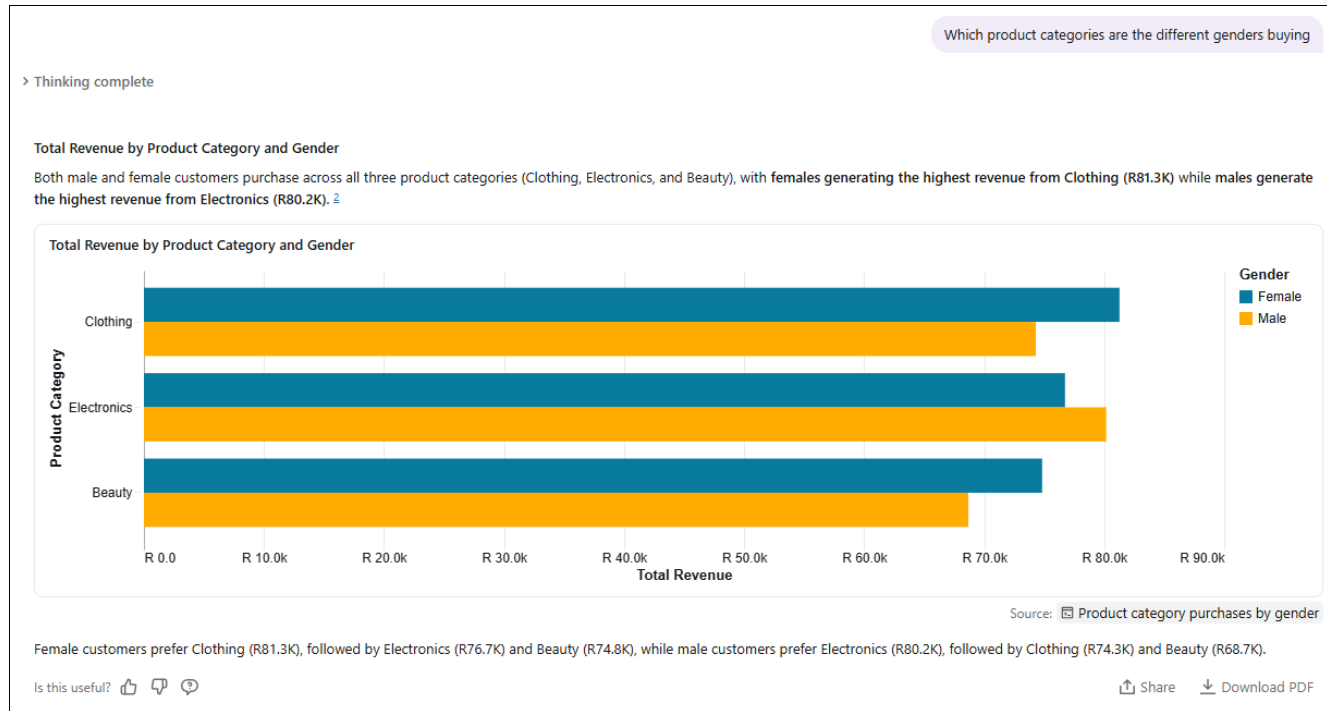
← Results 2 of 2 → Table ▾ +

	Product Category	total_revenue	avg_price	transaction_count	total_quantity
1	Electronics	156905	181.900584795321...	342	849
2	Clothing	155580	174.287749287749...	351	894
3	Beauty	143515	184.055374592833...	307	771

The agent's answer is correct.

Q7. The agent's original answer

Which product categories are the different genders buying



The supporting evidence you used to check it

- SQL query

```
18 SELECT
19   `Gender`,
20   `Product Category`,
21   COUNT(DISTINCT `Transaction ID`) as transaction_count,
22   SUM(`Total Amount`) as total_revenue
23 FROM
24   `workspace`.`data_agent`.`retail_sales_dataset`
25 WHERE
26   `Gender` IS NOT NULL
27   AND `Product Category` IS NOT NULL
28 GROUP BY
29   `Gender`,
30   `Product Category`
31 ORDER BY
32   `Gender`,
33   total_revenue DESC;
```

Add parameter

← Results 3 of 3 → Table +

	^A _C Gender	^A _C Product Category	¹ ₃ transaction_count	¹ ₃ total_revenue
1	Female	Clothing	174	81275
2	Female	Electronics	170	76735
3	Female	Beauty	166	74830
4	Male	Electronics	172	80170
5	Male	Clothing	177	74305
6	Male	Beauty	141	68685

The agent's answer is correct.

Key insights

- The business generated a total revenue of R456,000 across all transactions in 2023.
- Product category (Beauty) with high average price produces low revenue.
- The month of May has the highest revenue (R53.2K), while September has the lowest revenue (R23.6K).
- Females generate higher revenue and they prefer to buy Clothing products, while males generate lower revenue and they prefer to buy Electronics; however the AOV for both genders is similar (female = R456.5 and male = R455.4).
- Customers aged 43 generates highest revenue (R18.0K), followed by age 34 (R16.8K) and 51 (R16.1K).
- Distribution of revenue is similar across all product categories, with Electronics contributing 34.41%, followed by Clothing at 34.12% and Beauty at 31.47%.

Business recommendations

- Have discounts and special offers during September to improve sales in the lowest-performing month.
- Make promotions on gender specific Beauty products, to increase revenue for Beauty products.
- Introduce discounts and campaigns to attract the low-performing ages.
- Maintain a balanced inventory across Electronics, Clothing, and Beauty since all categories contribute similarly to total revenue.

Conclusion

It was easy to complete the steps 1-4 and 6-10. The challenge was step 5 (writing the instructions), but this will improve with more practice. The Data Agent was able to keep focused on the provided dataset only and it followed the instructions provided to it when answering questions. The challenge that the Data Agent had was interpreting the time analysis chart.